More on Specification and Data Issues

Intermediate Econometrics

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Wooldridge, Chapter 9

Outline

Functional Form Misspecification (§9-1)

Where We Are

Chapter 8 dealt with one Gauss–Markov failure: **heteroskedasticity**. This was a mild problem: OLS remained **unbiased and consistent**, only inference needed adjustment.

> **Chapter 9 returns to a much more serious problem: correlation between u and one or more explanatory variables** ($\text{Cov}(x_j, u) \neq 0$). This causes **bias and inconsistency in all OLS estimators**.

Three main reasons x_j can be endogenous (this chapter):

1. **Functional form misspecification** (§9-1) — omitting nonlinear terms
2. **Omitted variables** (§9-2) — addressed via *proxy variables*
3. **Measurement error** (§9-4) — biases of known sign

Plus three **data problems**:

- Missing data (§9-5a)
- Nonrandom sampling (§9-5b)
- Outliers and the LAD alternative (§9-5c, §9-6)

Functional Form Misspecification: Motivation & Setup

Motivation. Even with *every* relevant variable collected, OLS can be biased if y is related to the x_j through the **wrong functional form** — a missing quadratic, log, or interaction. We *have* the data; only the specification is wrong.

Setup. If the true model carries a nonlinear term we drop, that term is absorbed into the error and induces $\mathrm{Cov}(x_j, u) \neq 0$:

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + \underbrace{\gamma\, x_1^2 + \cdots}_{\text{dropped} \to u}.$$

Plan: (i) what misspecification *is*, (ii) detect it — F test on quadratics and the **RESET** test, (iii) choose between **nonnested** models.

Functional Form Misspecification: Definition

Functional Form Misspecification

A multiple regression suffers from **functional form misspecification** when it does not properly account for the relationship between the dependent variable and the observed explanatory variables.

Example: A Wage Equation

True model:

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 exper^2 + u.$$

If we estimate

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + u$$

(omitting $exper^2$), we have functional form misspecification.

Consequence: Generally biased estimates of β_0, β_1, β_2. The **return to experience** is no longer β_2 but $\beta_2 + 2\beta_3 exper$, so a single number cannot capture it.

Another Example: Missing Interactions

Consider the wage equation:

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 exper^2 + \beta_4 female + \beta_5 (\mathit{female} \cdot \mathit{educ}) + u.$$

If we **omit the interaction** *female · educ*:

- We **misspecify** the functional form.
- Estimators of *all* other parameters are biased.
- The ”return to education” is unclear because it depends on **gender**.

> **Key insight:** Functional form misspecification = the *wrong functional form* of *observed* variables. We **have** the data; we are just not using it correctly.
>
> Contrast this with **omitted variables** (§9-2), where data on a key variable *cannot be collected*.

Detecting Misspecification: The *F* Test for Quadratics

We already have a tool: the *F* **test for joint exclusion restrictions**.

Procedure:

1. Add **quadratic terms** for any significant explanatory variable.
2. Perform a joint *F* test of significance.
3. If significant: keep the quadratics (at the cost of complicating interpretation).

> **Symptomatic warning:** Significant quadratic terms can also indicate other functional form problems — e.g., using the **level** of a variable when the **logarithm** would be more appropriate, or vice versa.
>
> Adding quadratics + using logarithms covers many nonlinear relationships in economics.

Example 9.1: Economic Model of Crime (CRIME1)

Dependent variable: $narr86$ (number of arrests in 1986). $n = 2{,}725$.

Variable	(1) Linear	(2) With quadratics
$pcnv$	-0.133 (0.040)	0.553 (0.154)
$pcnv^2$	—	-0.730 (0.156)
$ptime86$	-0.041 (0.009)	0.287 (0.094)
$ptime86^2$	—	-0.0296 (0.0039)
$inc86$	-0.0015 (0.0003)	-0.0034 (0.0008)
$inc86^2$	—	-0.000007 (0.000003)
R^2	0.0723	0.1035

Joint F test on the three squared terms: $F = 31.37$ ($df = 3, 2713$), p-value ≈ 0. The squared terms are **jointly very significant**.

Interpretation: The initial linear model overlooked important nonlinearities. But interpreting the quadratics is harder: e.g., $pcnv$ has a positive deterrent effect only up to $pcnv = 0.365$, then negative.

The RESET Test

Adding quadratics is one approach but uses up many degrees of freedom. A more general test: **Ramsey's RESET (1969)**.

RESET Idea

Idea: if the original model

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + u$$

satisfies MLR.4, then **no nonlinear function of the** x_j should be significant when added.

RESET adds polynomials in the OLS fitted values $\hat{y}$ to detect general forms of misspecification. Squared and cubed terms are commonly used.

Why $\hat{y}$ instead of all the x_j's? $\hat{y}$ summarizes *all* the regressors in one variable, saving degrees of freedom.

The RESET Statistic

Let $\hat{y}$ be OLS fitted values from the original model. Estimate the **augmented regression**:

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + \delta_1 \hat{y}^2 + \delta_2 \hat{y}^3 + error.$$

RESET test:

$$\mathrm{H}_0 : \ \delta_1 = 0, \ \delta_2 = 0 \quad \text{(model correctly specified)}$$

$$F \sim F_{2,\, n-k-3} \quad \text{(approximately, in large samples).}$$

A **significant** F statistic suggests functional form misspecification.

Notes:

- An LM version is also available (chi-square distribution, 2 df).
- Both versions are easily made robust to heteroskedasticity (Ch. 8 methods).

Example 9.2: Housing Price Equation (HPRICE1, $n = 88$)

Model 1 (levels):

$$price = \beta_0 + \beta_1 lotsize + \beta_2 sqrft + \beta_3 bdrms + u.$$

Model 2 (log–log):

$$\log(price) = \beta_0 + \beta_1 \log(lotsize) + \beta_2 \log(sqrft) + \beta_3 bdrms + u.$$

	RESET F statistic	p-value
Model 1 (levels)	4.67	**0.012**
Model 2 (log–log)	2.56	0.084

Verdict:

- Reject Model 1 at the 5% level $\Rightarrow$ misspecification.
- Do not reject Model 2 at the 5% level (would reject at 10%).
- **The log–log model is preferred** based on RESET.

Cautions about RESET

What RESET can do:

- Detect general functional form misspecification (conditional mean).
- Compare two non-nested specifications informally.

What RESET cannot do:

- Detect **omitted variables** when those variables have linear conditional expectations in the included regressors.
- Detect **heteroskedasticity** (RESET is for the conditional *mean*).

Bottom line: RESET is a functional form test, **nothing more**.

Practical limitation: If RESET rejects, it does *not* tell you what to do next. You must investigate further (try logs, add interactions, use quadratics, etc.).

Tests against Nonnested Alternatives (§9-1b)

Some specifications are not nested in each other. Consider:

$$\text{Model A:} \quad y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u \tag{9.6}$$

$$\text{Model B:} \quad y = \beta_0 + \beta_1 \log(x_1) + \beta_2 \log(x_2) + u \tag{9.7}$$

Nonnested Models

Two models are **nonnested** if neither can be obtained from the other by imposing parameter restrictions. (Here: levels vs. logs.)

Cannot use a standard *F* test. Two approaches:

1. **Comprehensive (Mizon–Richard 1986):** construct a model nesting both, test restrictions.
2. **Davidson–MacKinnon (1981):** an auxiliary *t* test (next slide).

The Davidson–MacKinnon Test

Idea: If Model A is correct, then fitted values from the *other* model (Model B) should be **insignificant** when added to Model A.

Procedure (testing A against B):

1. Estimate Model B by OLS, save fitted values $\check{y}$.
2. Run the auxiliary regression:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \theta_1 \check{y} + \mathit{error}.$$

3. ***t*-statistic on θ_1** is the Davidson–MacKinnon statistic.

Decision: A significant *t*-statistic (two-sided alternative) is **evidence against Model A**.

To test Model B against Model A: swap roles. Save fitted values $\hat{y}$ from A, add to B as a regressor.

Caveats of Nonnested Testing

Three problems with nonnested tests:

1. **No clear winner needed:** Both models could be rejected, or neither rejected. If both rejected: more work needed. If neither: report both, use adjusted R^2 or theoretical considerations.
2. **Rejecting A does not validate B:** Model A can be rejected for any number of reasons, not just the one B captures.
3. **Different dependent variables:** Comparing y vs. $\log(y)$ requires additional adjustments. Goodness-of-fit comparisons need care (Wooldridge 1994a).

Proxy Variables for Unobserved Variables (§9-2)

Proxy Variables: Motivation & Setup

Motivation. The most damaging specification error is a **missing determinant we cannot observe** — e.g. *ability* in a wage equation. If it correlates with a regressor, OLS is biased and inconsistent (omitted-variable bias). Can we do anything *without* the variable itself?

Setup. We cannot drop the unobservable (that *is* the bias) and we cannot measure it. The compromise: substitute an observed **proxy** that is correlated with it. **Plan:** the proxy idea, the plug-in regression, the conditions under which it removes the bias, and using a **lagged dependent variable** as a proxy.

Omitted Variables: A Bigger Problem

Functional form misspecification is mild: we have all the data, just need the right form.

Omitted variables are harder: a key variable is unobserved, often because data *cannot be collected*.

Wage equation example:

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 abil + u$$

where $abil$ = ability — **unobservable**. If $educ$ is correlated with $abil$, putting $abil$ in the error causes **omitted variable bias**.

Question: Can we mitigate this bias if we cannot directly observe $abil$?
Answer: Sometimes — using a **proxy variable**.

The Proxy Variable Idea

Proxy Variable

A **proxy variable** is something *related to* the unobserved variable, used as a **substitute** in the regression.

Example: **IQ score** as a proxy for *ability*.

- Does *not* require IQ $=$ ability.
- Requires IQ to be **correlated** with ability.

Setup: Consider the model with three explanatory variables (two observed):

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3^* + u, \tag{9.10}$$

where x_3^* is **unobserved** but we have a proxy x_3:

$$x_3^* = \delta_0 + \delta_3 x_3 + v_3. \tag{9.11}$$

The intercept δ_0 allows x_3^* and x_3 to have different scales (e.g., ability has no natural scale, but IQ has mean 100).

The Plug-in Solution

Plug-in Solution to Omitted Variables

Pretend that x_3 and x_3^* are the same. Run:

$$y \text{ on } x_1,\ x_2,\ x_3. \tag{9.12}$$

Use x_3 as a **stand-in** for the unobservable x_3^*.

When does this work? Two assumptions are sufficient:

1. u **uncorrelated with** $x_1,\ x_2,\ x_3^*,\ x_3$**:** The first three are standard. The new requirement: u uncorrelated with x_3. Holds if x_3 is **irrelevant** once x_3^* is controlled for.
2. v_3 **uncorrelated with** $x_1,\ x_2,\ x_3$**:** Means x_3 is a **good proxy** for x_3^*:

$$\mathbb{E}(x_3^* \mid x_1, x_2, x_3) = \mathbb{E}(x_3^* \mid x_3) = \delta_0 + \delta_3 x_3. \tag{9.13}$$

Why the Plug-in Solution Works

Plug the proxy equation $x_3^* = \delta_0 + \delta_3 x_3 + v_3$ into the model:

$$y = (\beta_0 + \beta_3\delta_0) + \beta_1 x_1 + \beta_2 x_2 + \beta_3\delta_3 x_3 + (u + \beta_3 v_3).$$

Define the composite error $e = u + \beta_3 v_3$. Then:

$$y = \alpha_0 + \beta_1 x_1 + \beta_2 x_2 + \alpha_3 x_3 + e,$$

where $\alpha_0 = \beta_0 + \beta_3\delta_0$ and $\alpha_3 = \beta_3\delta_3$.

What we get:

- **Unbiased estimates of β_1 and β_2** (the parameters of interest).
- **Unbiased estimate of $\alpha_3 = \beta_3\delta_3$** (interpretable as the partial effect of the proxy on y).
- **Cannot recover β_3 alone** (we do not observe δ_3).

Often α_3 is more interesting anyway: e.g., the wage premium per IQ point is more directly useful than the wage premium per ”unit of ability.”

Example 9.3: IQ as a Proxy for Ability (WAGE2)

Data: WAGE2 (Blackburn–Neumark 1992), $n = 935$ men. Includes monthly earnings, education, demographics, and IQ scores from 1980.

	(1) No IQ	(2) With IQ	(3) With IQ + interaction
$educ$	0.065 (0.006)	0.054 (0.007)	0.053 (0.007)
$exper$	0.014 (0.003)	0.014 (0.003)	0.014 (0.003)
$tenure$	0.012 (0.002)	0.011 (0.002)	0.011 (0.002)
IQ	—	0.0036 (0.0010)	0.0036 (0.0010)
$(educ - \overline{educ})(IQ - \overline{IQ})$	—	—	0.00034 (0.00038)
R^2	0.253	0.263	0.263

Key finding:

- Without IQ: estimated return to education = **6.5%**.
- With IQ as proxy: return falls to **5.4%**.
- The drop $(1.1$ percentage points) corresponds to omitted ability bias.
- IQ premium: 3.6% per 10 IQ points (since SD of IQ ≈ 15, a one-SD increase $\Rightarrow$ 5.4% higher earnings).

The interaction $(educ \cdot IQ)$ in column (3) is insignificant ($t \approx 0.89$): no convincing evidence of an interactive effect.

When the Proxy Is Imperfect

Suppose the proxy depends on the other regressors:

$$x_3^* = \delta_0 + \delta_1 x_1 + \delta_2 x_2 + \delta_3 x_3 + v_3. \tag{9.14}$$

Then plugging in:

$$y = (\beta_0 + \beta_3\delta_0) + (\beta_1 + \beta_3\delta_1)x_1 + (\beta_2 + \beta_3\delta_2)x_2 + \beta_3\delta_3 x_3 + (u + \beta_3 v_3).$$

Bias formula:

$$\text{plim}(\hat{\beta}_1) = \beta_1 + \beta_3\delta_1, \qquad \text{plim}(\hat{\beta}_2) = \beta_2 + \beta_3\delta_2.$$

Bias remains, but **should be smaller** than ignoring the omitted variable entirely.

Wage example: If $\delta_1 > 0$ (more education $\Rightarrow$ higher ability) and $\beta_3 > 0$ (ability $\Rightarrow$ higher wage), then $\hat{\beta}_1$ remains upward biased *even with IQ as proxy*. But less than without IQ.

Lagged Dependent Variables as Proxies (§9-2a)

Sometimes we do not even have a candidate proxy. A useful alternative: **the lagged dependent variable**.

Example 9.4: City Crime Rates

$$crime = \beta_0 + \beta_1 unem + \beta_2 expend + \beta_3 crime_{-1} + u. \tag{9.16}$$

- $crime_{-1}$ = crime rate from an earlier year.
- $\beta_3 > 0$ expected: crime has **inertia**.
- Including $crime_{-1}$ controls for **historical unobservables** that drive both past and current crime.

Reading β_2: The effect on crime of an extra dollar of law enforcement, **holding constant prior crime rate**. This is a much cleaner causal interpretation than a pure cross-sectional regression.

Crime Rate Estimates (CRIME2, $n = 46$ cities)

Variable	(1) No lag	(2) With lagged crime
unem$_{87}$	-0.029 (0.032)	0.009 (0.020)
$\log($*lawexpc*$_{87})$	0.203 (0.173)	-0.140 (0.109)
$\log($*crmrte*$_{82})$	—	1.194 (0.132)
intercept	3.34 (1.25)	0.076 (0.821)
R^2	0.057	0.680

Findings:

- **Without** lagged crime: counterintuitive — more law enforcement spending seems to *increase* crime ($t \approx 1.17$).
- **With** lagged crime: elasticity of crime w.r.t. enforcement becomes -0.14 ($t \approx -1.28$).
- Lagged crime is highly significant (elasticity ≈ 1): cities with high past crime tend to have high current crime.

Why the original sign was wrong: Cities with high recent crime spend more on law enforcement, creating reverse correlation. Lagged crime ”soaks up” this confounding.

Why Was Unemployment Negative Without the Lag?

Theory predicts $\beta_{unem} \geq 0$ (joblessness raises crime), yet column (1) reports -0.029. Two points resolve the puzzle.

1. It is not significant. $t = -0.029/0.032 \approx -0.9$ — the negative estimate is **statistically indistinguishable from zero**. The "wrong sign" is sampling noise, not a finding: never over-read a single insignificant coefficient.

2. Omitted-variable bias. Column (1) leaves out **persistent city traits** (demographics, socioeconomic conditions, reporting practices) that drive crime; $\log(crmrte_{82})$ proxies for them. With δ = effect of those traits on crime and γ = their partial association with $unem$,

$$\operatorname{plim} \hat{\beta}^{(1)}_{unem} = \beta_{unem} + \delta\,\gamma.$$

The coefficient *rises* ($-0.029 \to +0.009$) once the lag is added, so the bias $\delta\gamma$ was **negative**. Since $\delta > 0$ (past crime enters at $+1.19$), we infer $\gamma < 0$: in this sample, higher-baseline-crime cities had *lower* 1987 unemployment.

After the lag: the confounding is absorbed and $\hat{\beta}_{unem}$ returns to its expected non-negative sign ($+0.009$) — though still insignificant at $n = 46$.

A Different Slant on Multiple Regression (§9-2b)

Alternative interpretation of regression with proxies:

> Instead of seeing IQ as a proxy for the latent concept ”ability,” interpret the wage equation more modestly:
>
> *”Among workers with the same IQ, experience, etc., what is the expected difference in log wage from one more year of education?”*

Predictive perspective:

- Sometimes the goal is **prediction**, not causal inference.
- Then ”omitted variable bias” loses its meaning.
- Focus instead on building a good predictive model with available regressors.
- Do not include variables **unobserved at the time of prediction**.

Example: A college admissions office uses high school GPA, SAT, etc. to predict college GPA. They are not concerned about omitted variable bias from omitting ”motivation.”

Potential Outcomes and Proxy Variables (§9-2c)

The proxy-variable idea links directly to the **potential outcomes** framework (Ch. 2–7). For a binary treatment $w \in \{0, 1\}$ with potential outcomes $y(0), y(1)$:

$$y(0) = \mu_0 + v(0), \qquad y(1) = \mu_1 + v(1), \qquad \tau_{ate} = \mu_1 - \mu_0.$$

Selection problem: participation w may be correlated with the unobservables $v(0), v(1)$. **Unconfoundedness (ignorability):** conditional on covariates $\mathbf{x}$, w is independent of $\big(y(0), y(1)\big)$ — which is *exactly the proxy-variable condition*: $\mathbf{x}$ proxies for the unobserved factors driving both the outcome and the treatment decision.

Estimation. With linear forms $\mathbb{E}[v(g) \mid w, \mathbf{x}] = \mathbb{E}[v(g) \mid \mathbf{x}] = (\mathbf{x} - \boldsymbol{\eta})\boldsymbol{\beta}_g$, $\boldsymbol{\eta} = \mathbb{E}(\mathbf{x})$, the ATE comes from one interacted regression over the *entire* sample,

$$y_i \text{ on } w_i,\ \mathbf{x}_i,\ w_i \cdot (\mathbf{x}_i - \bar{\mathbf{x}}),$$

where the coefficient on w_i is $\hat{\tau}_{ate}$ (derived next, §7-6).

The Parameter of Interest and the ATE Regression (§7-6)

Setup. Each unit has potential outcomes $y(1)$ (if treated) and $y(0)$ (if untreated); with treatment $w \in \{0, 1\}$ we observe only one, $y = (1 - w)\, y(0) + w\, y(1)$, where $y(g) = \mu_g + v(g)$ and $\mu_g = \mathbb{E}[y(g)]$.

Parameter of interest — the average treatment effect (ATE):

$$\tau \;=\; \mathbb{E}[\, y(1) - y(0)\,] \;=\; \mu_1 - \mu_0.$$

The *unit-level* effect $y_i(1) - y_i(0)$ is never observed, so we target its average.

Derivation. Take $\mathbb{E}(\cdot \mid w, \mathbf{x})$; **unconfoundedness** drops w from each conditional mean:

$$\begin{aligned}\mathbb{E}(y \mid w, \mathbf{x}) &= \mathbb{E}[y(0) \mid \mathbf{x}] + w\left\{\mathbb{E}[y(1) \mid \mathbf{x}] - \mathbb{E}[y(0) \mid \mathbf{x}]\right\} \\ &= \mu_0 + (\mathbf{x} - \boldsymbol{\eta})\boldsymbol{\beta}_0 + \underbrace{(\mu_1 - \mu_0)}_{\tau}\, w + (\boldsymbol{\beta}_1 - \boldsymbol{\beta}_0)\, w(\mathbf{x} - \boldsymbol{\eta}).\end{aligned}$$

$\Rightarrow$ regress y on w, $\mathbf{x}$, $w(\mathbf{x} - \bar{\mathbf{x}})$ ($\boldsymbol{\eta} = \mathbb{E}(\mathbf{x})$ replaced by $\bar{\mathbf{x}}$); the coefficient on w estimates τ.

Why That Form? Interaction and Centering (§7-6)

Why the interaction $w(\mathbf{x}-\bar{\mathbf{x}})$**?** It lets the treated and control groups have *different slopes* ($\boldsymbol{\beta}_1 \neq \boldsymbol{\beta}_0$), so the effect may vary with covariates:

$$\tau(\mathbf{x}) = \tau + (\mathbf{x} - \boldsymbol{\eta})(\boldsymbol{\beta}_1 - \boldsymbol{\beta}_0).$$

Dropping it forces a *constant* effect $\tau(\mathbf{x}) \equiv \tau$.

Why *center* at $\bar{\mathbf{x}}$**?** The ATE is the *average* of $\tau(\mathbf{x})$, and $\mathbb{E}(\mathbf{x} - \boldsymbol{\eta}) = 0$, so the coefficient on *w* equals τ **only when the interaction is demeaned**. With a raw $w\mathbf{x}$ term, that coefficient would instead be the effect at $\mathbf{x} = \mathbf{0}$ — a (usually fictional) unit, not the population average.

Models with Random Slopes (§9-3)

Models with Random Slopes: Motivation & Setup

Motivation. Every model so far forces *one* slope β_j on the whole population. But the return to education, or the effect of a price change, plausibly **differs across individuals** by unobserved traits. What is OLS estimating when slopes are heterogeneous?

Setup. Allow each unit its own intercept a_i and slope b_i, and define the population averages $\alpha = \mathbb{E}(a_i)$ and $\beta = \mathbb{E}(b_i)$, so that β is the **average partial effect (APE)**. **Plan:** the conditions under which OLS still recovers α, β, the heteroskedasticity it induces, and when it breaks down.

The Random Slopes Model

Until now, we have assumed all individuals share the same slope β_j. What if the slope varies across the population by **unobservable** characteristics?

Random Coefficient (Random Slope) Model

For unit i in the population:

$$y_i = a_i + b_i x_i, \tag{9.17}$$

where:

- a_i = unit-specific **intercept**
- b_i = unit-specific **slope**
- Both vary across the population

Example: Wage equation with b_i = individual return to education. Different people benefit differently from another year of schooling depending on **unobserved ability, motivation,** etc.
Define $\alpha = \mathbb{E}(a_i)$ and $\beta = \mathbb{E}(b_i)$ — the **average intercept** and **average partial effect (APE)**.

When OLS Still Works

Write $a_i = \alpha + c_i$, $b_i = \beta + d_i$ with $\mathbb{E}(c_i) = 0$, $\mathbb{E}(d_i) = 0$. Substituting into (9.17):

$$y_i = \alpha + \beta x_i + (c_i + d_i x_i) = \alpha + \beta x_i + u_i, \tag{9.18}$$

where the new error is $u_i = c_i + d_i x_i$.

Sufficient conditions for OLS to estimate α and β unbiasedly:

$$\mathbb{E}(a_i \mid x_i) = \mathbb{E}(a_i), \qquad \mathbb{E}(b_i \mid x_i) = \mathbb{E}(b_i). \tag{9.19}$$

That is: a_i and b_i are **mean independent** of x_i.

Then OLS recovers the **population averages** of the unit-specific slopes.

Inevitable side effect: The error $u_i = c_i + d_i x_i$ is **heteroskedastic**:

$$\mathrm{Var}(u_i \mid x_i) = \sigma_c^2 + \sigma_d^2 x_i^2. \tag{9.20}$$

Use **heteroskedasticity-robust standard errors** (Ch. 8).

Random Slopes: Practical Takeaway

Key result: If we just want to estimate the **average partial effect** of x on y (the APE), and a_i, b_i are independent of x_i, then **OLS works**.

- Use OLS as before.
- Always use heteroskedasticity-robust standard errors.
- Interpretation changes: $\hat{\beta}$ is the **average** return, not the return for any specific person.

When OLS fails: If b_i is correlated with x_i (e.g., people who benefit more from education *select into* more education), OLS is biased. Solution: instrumental variables (Ch. 15).

Connection to Ch. 7–8: Random slopes provide **another justification** for using interaction terms or heteroskedasticity-robust SEs in standard regression analysis.

Properties of OLS under Measurement Error (§9-4)

Measurement Error: Motivation & Setup

Motivation. We often record a variable *with error*: reported vs true income, average vs marginal tax rate, self-reported vs actual hours. Does a mismeasured variable bias OLS — and if so, in **which direction**?

Setup. Write each observed value as truth plus noise, e.g. $x_1 = x_1^* + e_1$, and trace how the noise e_1 propagates into $\hat{\beta}$. **Plan:** (i) measurement error in the *dependent* variable (usually harmless), then (ii) error in an *explanatory* variable under the **classical errors-in-variables** assumption $\Rightarrow$ **attenuation bias** toward zero.

Measurement Error: The Issue

Sometimes we cannot collect data on the variable that truly affects economic behavior.

Example: Marginal Tax Rate

A family chooses charitable contributions based on their **marginal income tax rate**. But the marginal rate depends on income brackets and is hard to compute. Researchers often use **average tax rate** as a substitute — measurement error.

Measurement Error vs. Proxy Variables

Both involve a substitute for an unobserved variable, but:

- **Proxy:** the unobserved variable is *vague* (e.g., ”ability”); the proxy (e.g., IQ) is just *associated* with it.
- **Measurement error:** the unobserved variable has a *well-defined, quantitative meaning*; we just cannot measure it precisely.

This section: **when does measurement error cause problems for OLS?**

Measurement Error in the Dependent Variable (§9-4a)

Let y^* be the true variable, y be the observed (measured) variable. The **measurement error** is

$$e_0 = y - y^*, \tag{9.24}$$

so $y = y^* + e_0$. Substituting into the true model:

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + (u + e_0). \tag{9.25}$$

Standard assumption: e_0 is **independent of** the x_j's.

Under this assumption:

- $\mathbb{E}(e_0) = 0$ (or absorbed in intercept).
- OLS is **unbiased and consistent.**
- Inference (t, F, LM tests) is valid.
- **Only cost:** larger error variance $\Rightarrow$ larger SEs.

Bottom line: Random measurement error in y is innocuous — it only inflates standard errors. Cannot fix without better data.

Example 9.5: Savings Function

Consider a savings function:

$$sav^* = \beta_0 + \beta_1 inc + \beta_2 size + \beta_3 educ + \beta_4 age + u.$$

- Observed savings sav may differ from true sav^* due to reporting error.
- **If** measurement error is uncorrelated with inc, $size$, $educ$, age — OLS is fine.
- **But** families with higher income or more education may report savings *more accurately* — measurement error correlated with regressors $\Rightarrow$ bias.

Logarithmic case: If the dependent variable is $\log(y^*)$:

$$\log(y) = \log(y^*) + e_0,$$

which corresponds to a **multiplicative** measurement error in y: $y = y^* a_0$ with $e_0 = \log(a_0)$. Common in firm-level data, where reported figures may be a multiplicative fraction of true values.

Measurement Error in an Explanatory Variable (§9-4b)

Now the **important** case. Let x_1^* be true, x_1 observed:

$$e_1 = x_1 - x_1^*. \tag{9.28}$$

Standard assumptions:

- $\mathbb{E}(e_1) = 0$.
- u uncorrelated with x_1^* *and* x_1 (i.e., once x_1^* is controlled for, x_1 has no separate effect on y).

Two Polar Cases

1. **e_1 uncorrelated with the *observed* x_1:**

$$\mathrm{Cov}(x_1, e_1) = 0. \tag{9.29}$$

   Like proxy variable case. OLS is consistent (with larger variance).

2. **Classical Errors-in-Variables (CEV):**

$$\mathrm{Cov}(x_1^*, e_1) = 0. \tag{9.30}$$

   The error is uncorrelated with the **true** variable. **Standard textbook assumption**.

The Attenuation Bias under CEV

Under CEV: $x_1 = x_1^* + e_1$ with $\mathrm{Cov}(x_1^*, e_1) = 0$. Then:

$$\mathrm{Cov}(x_1, e_1) = \mathrm{Cov}(x_1^* + e_1, e_1) = \sigma_{e_1}^2.$$

Substituting $x_1^* = x_1 - e_1$ into $y = \beta_0 + \beta_1 x_1^* + u$:

$$y = \beta_0 + \beta_1 x_1 + (u - \beta_1 e_1). \tag{9.32}$$

The composite error $u - \beta_1 e_1$ is **correlated with** x_1. Hence OLS is biased.

The bias formula:

$$\mathrm{plim}(\hat{\beta}_1) = \beta_1 \cdot \frac{\sigma_{x_1^*}^2}{\sigma_{x_1^*}^2 + \sigma_{e_1}^2}. \tag{9.33}$$

The factor $\frac{\sigma_{x_1^*}^2}{\sigma_{x_1^*}^2 + \sigma_{e_1}^2}$ is between 0 and 1.

Conclusion: $\hat{\beta}_1$ is **always closer to zero** than β_1. This is the **attenuation bias**.

How Big Is the Attenuation?

Define the **signal-to-noise ratio** $r = \sigma^2_{x_1^*}/\sigma^2_{e_1}$.

$$\text{plim}(\hat{\beta}_1) = \beta_1 \cdot \frac{r}{r+1}.$$

Signal-to-noise r	Attenuation factor	Bias if $\beta_1 = 1$
$r = 1$ (50% noise)	0.50	$\hat{\beta}_1 \to 0.50$
$r = 4$ (20% noise)	0.80	$\hat{\beta}_1 \to 0.80$
$r = 9$ (10% noise)	0.90	$\hat{\beta}_1 \to 0.90$
$r = 99$ (1% noise)	0.99	$\hat{\beta}_1 \to 0.99$

Implications:

- Sign of $\hat{\beta}_1$ is preserved (only magnitude shrinks).
- If $\beta_1 > 0$, $\hat{\beta}_1$ **understates** the true effect.
- Hypothesis test of $H_0 : \beta_1 = 0$ has **less power**.
- More variation in x_1^* relative to $e_1 \Rightarrow$ less bias.

Multiple Regression with CEV

Suppose only x_1^* is mismeasured:

$$y = \beta_0 + \beta_1 x_1^* + \beta_2 x_2 + \beta_3 x_3 + u, \tag{9.34}$$

with x_2, x_3 measured perfectly. Under CEV:

Bias on the mismeasured variable:

$$\operatorname{plim}(\hat{\beta}_1) = \beta_1 \cdot \frac{\sigma^2_{r_1^*}}{\sigma^2_{r_1^*} + \sigma^2_{e_1}}, \tag{9.36}$$

where r_1^* is the **population residual** from regressing x_1^* on x_2, x_3.

Bias on the other coefficients: Generally, $\hat{\beta}_2$ and $\hat{\beta}_3$ are **also inconsistent** (due to indirect correlation). Sizes and signs of these biases are hard to predict.

Bottom line: Measurement error in *one* regressor contaminates *all* OLS estimates.

Example 9.7: GPA Equation with Measurement Error

Suppose true model:

$$colGPA = \beta_0 + \beta_1 faminc^* + \beta_2 hsGPA + \beta_3 SAT + u,$$

where $faminc^*$ = actual annual family income (mismeasured if students self-report).

Observed: $faminc = faminc^* + e_1$.

If CEV holds:

- OLS estimate of β_1 is **biased toward zero**.
- Reduced power for $H_0 : \beta_1 = 0$.
- Practical fear: cannot reject "family income has no effect on college GPA" even if the true effect is nontrivial.

Solution preview: If a second *noisy measurement* of family income is available (or another instrument), **instrumental variables** (Ch. 15) can correct the attenuation bias.

Missing Data, Nonrandom Samples, Outliers (§9-5)

Data Problems: Motivation & Setup

Motivation. Real samples are imperfect: observations have **gaps**, the sample may be drawn **nonrandomly**, and a few **outliers** can dominate the fit. Each can threaten random sampling (MLR.2) — but not all of them bias OLS.

Setup. For every data problem the decisive question is the same: *is an observation's inclusion or exclusion related to the error u?*

- **Missing data** (§9-5a): MCAR / MAR / NMAR; complete-case vs. the missing-indicator method.
- **Nonrandom samples** (§9-5b): *exogenous* (harmless) vs. *endogenous* (biasing) selection.
- **Outliers** (§9-5c): influential observations — which motivate **LAD** in §9-6.

Missing Data (§9-5a)

Three Types of Missingness

1. **Missing Completely At Random (MCAR):** the probability of missingness is *independent of* all observed and unobserved factors affecting y.
2. **Missing At Random (MAR):** probability of missingness can depend on observed regressors but not on unobservables.
3. **Not Missing At Random (NMAR):** missingness depends on the unobserved value itself.

If MCAR holds: dropping observations with missing data (the **complete cases estimator**) is OLS-consistent. Just a smaller sample.

Example: If 196 of 1,388 observations in BWGHT are missing $faminc$, simply drop them. As long as missingness is unrelated to factors affecting y, the remaining sample is still random.

The Missing Indicator Method (MIM)

Suppose only x_k is sometimes missing. **MIM creates two new variables:**

- z_{ik}: equals x_{ik} when observed, **0 otherwise**.
- m_{ik}: a "missing indicator" — 1 if missing, 0 if observed.

Missing Indicator Method

Run the regression

$$y_i \text{ on } x_{i1}, \ldots, x_{i,k-1}, z_{ik}, m_{ik}, \; i = 1, \ldots, n.$$

Use the **full sample** of n observations.

Appeal: Uses all n observations rather than just the complete cases.

Catch: Statistically valid **only under strong assumptions** — essentially requires x_k uncorrelated with other regressors (Jones 1996; Abrevaya & Donald 2017). These conditions **rarely hold** in practice.

MCAR vs. MIM: Which to Use?

Property	Complete Cases (MCAR)	Missing Indicator Method
Sample size used	Smaller	Full n
Required assumption	MCAR	MCAR + $x_k \perp$ other regressors
Robustness	More robust	Less robust
Bias if assumption fails	Inconsistent	Inconsistent

Wooldridge's recommendation: The complete cases estimator is generally preferred because it requires **weaker assumptions**. MIM should be used cautiously — it can cause more harm than good if its assumptions fail.

More advanced: Multiple imputation, EM algorithm — beyond scope.

Nonrandom Samples (§9-5b)

Even without missing data, the **sample itself** may be nonrandomly selected.

Two Types of Sample Selection

1. **Exogenous sample selection:** Selection based on the *independent* variables.
   Example: We sample only people over 35 in a savings study.
2. **Endogenous sample selection:** Selection based on the *dependent* variable (or unobservables affecting y).
   Example: We sample only households with wealth $< \$250{,}000$.

Statistical consequence:

- Exogenous selection $\Rightarrow$ OLS is **still unbiased and consistent**.
- Endogenous selection $\Rightarrow$ OLS is **biased and inconsistent**.

Sample Selection: Examples

Exogenous Selection: Saving Study

$$saving = \beta_0 + \beta_1 income + \beta_2 age + \beta_3 size + u.$$

Sample only people over 35. Selection is on age (a regressor). **OLS is unbiased.** Just a smaller, less general sample.

Endogenous Selection: Wealth Equation

$$wealth = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 age + u.$$

Sample only households with *wealth* $< \$250{,}000$. Selection is on *y*. **OLS is biased.** The conditional expectation in the truncated population is not the same as in the full population.

Stratified sampling: If sampling weights vary by strata defined by **exogenous** variables, OLS on the unweighted data is consistent. If strata are defined by endogenous variables, special methods are needed.

Wage Equations and Working Samples

A subtle case: estimating wage equations using **only working individuals**.

$$\log(wage^o) = \beta_0 + \beta_1 educ + \beta_2 exper + u, \tag{9.39}$$

where $wage^o$ = wage *offer* (wage someone could earn if they worked).

- For non-workers, $wage^o$ is unobserved.
- We estimate using only those who chose to work.
- If the choice to work depends on **unobserved factors** affecting wage offers (motivation, family circumstances) $\Rightarrow$ **endogenous selection**.

Heckman's solution: The Heckman two-step procedure (Ch. 17) corrects for sample selection bias when the selection mechanism can be modeled.

Outliers and Influential Observations (§9-5c)

Outlier vs. Influential Observation

- **Outlier:** An observation with unusual y and/or x values relative to the rest of the sample.
- **Influential observation:** An outlier whose removal substantially *changes* the OLS estimates.

Why OLS is sensitive: OLS minimizes the *sum of squared* residuals. Large residuals are weighted heavily (u^2 grows quadratically), so single observations can dominate the fit.

Two sources of outliers:

1. **Data entry errors** — always check summary statistics (min, max).
2. **Genuine extreme cases** — small populations, mixtures of distributions.

Example 9.8: R&D Intensity and Firm Size (RDCHEM)

Data: 32 chemical companies, $rdintens$ = R&D as % of sales, $sales$ in millions.

All 32 firms:

$$\widehat{rdintens} = 2.625 + 0.000053\,sales + 0.0446\,profmarg,$$

with standard errors 0.586, 0.000044, 0.0462. $R^2 = 0.076$. **No coefficient is significant at 10%.**

Drop largest firm ($sales \approx \$40$ billion, twice the next largest):

$$\widehat{rdintens} = 2.297 + 0.000186\,sales + 0.0478\,profmarg,$$

with new standard errors 0.592, 0.000084, 0.0445. $R^2 = 0.173$.

Striking result: Coefficient on $sales$ **more than triples**, becomes **significant at 5%**. The single large firm was masking the relationship.

Detecting Outliers: Studentized Residuals

Are large residuals a reliable sign of outliers? **Not directly.**

> **Why raw residuals mislead:** OLS *fits* to outliers, so the residual at an outlier may actually be **small** (the line is dragged toward the outlier).

Studentized Residuals

The **studentized residual** for observation h is the OLS residual divided by an estimate of its standard deviation, computed **leaving observation h out**.

Easy computation trick: Add a dummy variable d_h for observation h to the full regression. The ***t*-statistic** on d_h *is* the studentized residual for h, with a t_{n-k-2} distribution.

Decision rule: A large absolute studentized residual ($|t| > 2$ or 3) flags observation h as an outlier.

Studentized Residuals: Caveats

In Example 9.8 (32 firms):

- Largest firm's studentized residual: -1.82 ($p = 0.08$). Marginally significant but not extreme.
- First observation (*rdintens* ≈ 9.42) has studentized residual 4.56 — much larger.

Surprise: Dropping the first observation *barely* changes the estimates. The largest studentized residual is **not** the most influential observation.

Lesson: Studentized residuals identify *extreme residuals*, not necessarily *influential observations*.

Best practice:

- Plot the data — visualize before computing.
- Run regression **with and without** suspect observations.
- Report differences honestly; let readers judge.
- **Do not just delete** unless the observation is clearly a data entry error.

Example 9.10: Infant Mortality (INFMRT)

$n = 51$ states + DC. All variables in logs:

$$\widehat{\textit{infmort}} = 33.86 - 4.68\log(pcinc) + 4.15\log(physic) - 0.088\log(popul).$$

Standard errors: 20.4, 2.6, 1.5, 0.29. $R^2 = 0.139$.

Surprise: more physicians per capita $\Rightarrow$ *higher* infant mortality?

DC is unusual: highest infant mortality (20.7 vs. 12.4 max elsewhere) AND most physicians (615 vs. 337 max elsewhere). Both extreme.

Drop DC:

$$\widehat{\textit{infmort}} = 23.95 - 0.57\log(pcinc) - 2.74\log(physic) + 0.629\log(popul).$$

Now $\log(physic)$ has the **expected negative sign** and is significant.

Conclusion: DC was **driving** the wrong sign. Reasonable to exclude (DC has unique demographic concentrations).

Functional Form and Outlier Sensitivity

Some functional forms are **less sensitive** to outliers.

Logarithmic transformations narrow the range of economic variables and reduce outlier influence.

Example 9.9 (RDCHEM in logs): Re-estimate the R&D model in log–log:

$$\log(rd) = \beta_0 + \beta_1 \log(sales) + \beta_2 profmarg + u.$$

All 32 firms:

$$\widehat{\log(rd)} = -4.378 + 1.084 \log(sales) + 0.0217\, profmarg,\ R^2 = 0.918.$$

Drop largest firm:

$$\widehat{\log(rd)} = -4.404 + 1.088 \log(sales) + 0.0218\, profmarg,\ R^2 = 0.904.$$

Practically identical results. The log–log specification is much more robust to outliers — a strong argument for using logs in skewed economic data.

Least Absolute Deviations Estimation (§9-6)

Least Absolute Deviations: Motivation & Setup

Motivation. OLS minimizes *squared* residuals, so a single extreme y can swing every coefficient. Rather than hunt down and delete outliers, can we estimate with a rule that is **intrinsically resistant** to them?

Setup. Replace OLS's quadratic loss with an **absolute-value** loss — minimize the sum of *absolute* residuals rather than squared ones. This estimates the **conditional median** instead of the mean. **Plan:** the LAD criterion, why the median is robust to outliers, and the robustness–efficiency trade-off against OLS.

Least Absolute Deviations (LAD)

Rather than identifying and removing outliers, an alternative: use an estimator **less sensitive** to outliers.

LAD Estimator

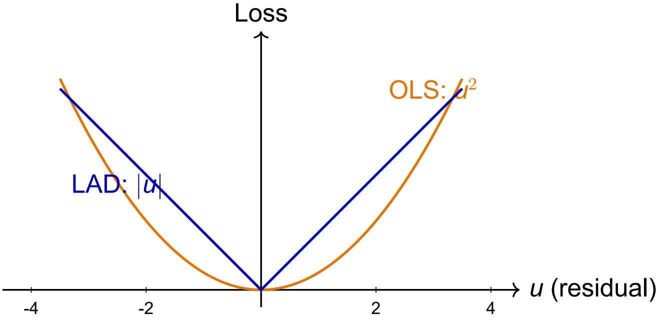
The LAD estimators of $\beta_0, \beta_1, \ldots, \beta_k$ **minimize the sum of absolute residuals**:

$$\min_{b_0, b_1, \ldots, b_k} \sum_{i=1}^{n} |y_i - b_0 - b_1 x_{i1} - \cdots - b_k x_{ik}|. \tag{9.45}$$

Key contrast with OLS:

- **OLS** minimizes $\sum (residual)^2$ — large residuals heavily penalized.
- **LAD** minimizes $\sum |residual|$ — outliers receive **linear** penalty, not quadratic.

LAD vs. OLS: Visual Comparison



Reading the picture:

- OLS loss grows **quadratically** — large residuals dominate.
- LAD loss grows **linearly** — large residuals get bounded weight.
- LAD is **robust to outliers** in y.

LAD Estimates the Conditional Median

LAD and the Conditional Median

OLS estimates the **conditional mean**: $\mathbb{E}(y \mid x_1, \ldots, x_k)$.

LAD estimates the **conditional median**: $\text{Med}(y \mid x_1, \ldots, x_k)$.

Why this matters: The median is not affected by extreme observations in the same way as the mean. So LAD estimates are **resilient** to outliers in y.

Symmetry caveat: If the conditional distribution of y given x's is **symmetric**, then **mean = median**, and LAD and OLS estimate the same parameters.

If asymmetric: LAD and OLS estimate *different* quantities. The choice depends on **whether you care about the mean or the median** effect.

LAD: Computation and Inference

Computation:

- No closed-form solution (unlike OLS).
- Requires **linear programming**.
- Historically computationally heavy; modern computers handle large samples easily.

Inference:

- No **exact** small-sample distributions (t, F).
- Statistical inference is **asymptotic** (large-sample).
- Software reports asymptotic standard errors based on the conditional density of the error at zero.
- Formulas are more complex than OLS — see Koenker (2005) for treatment.

Practical note: For modest samples ($n < 200$) the asymptotic approximation may be poor — bootstrap is often preferred.

When to Choose LAD over OLS

Scenario	Use OLS	Use LAD
No outliers, symmetric errors	✓ (more efficient)	—
Outliers in y	risky	✓ (robust)
Asymmetric errors	estimates mean	estimates median
Skewed dependent variable	risky	✓ (often more interpretable)
Need exact small-sample inference	✓	×
Large samples	✓	✓
Mean is the policy quantity	✓	×
Median is the policy quantity	×	✓

Recommendation: Always run **both OLS and LAD** as a robustness check on important results. Differences between them *point to specific problems* (outliers, asymmetry).

Quantile Regression Generalization

LAD estimates the **conditional median**. We can generalize to other quantiles.

Quantile Regression (Koenker–Bassett 1978)

For quantile $\tau \in (0,1)$, the τ-th quantile regression minimizes:

$$\min_{b_0,\dots,b_k} \sum_{i=1}^{n} \rho_\tau\left(y_i - b_0 - b_1 x_{i1} - \cdots - b_k x_{ik}\right),$$

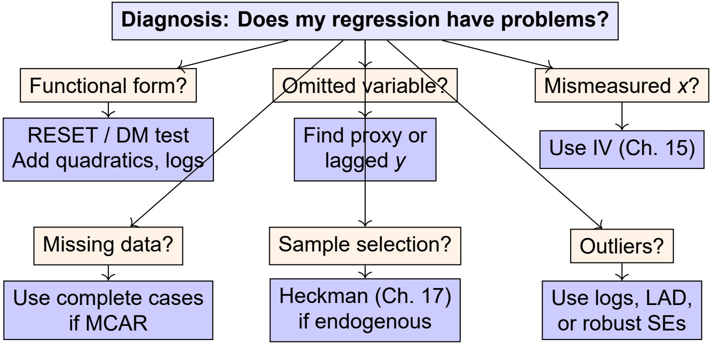
where $\rho_\tau(u) = u\left(\tau - \mathbf{1}\{u < 0\}\right)$ is the **check function**.

Special case: $\tau = 0.5$ gives LAD (median regression).

Why useful:

- Different effect at the **bottom** ($\tau = 0.1$) vs. **top** ($\tau = 0.9$) of the conditional distribution of *y*.
- E.g., effect of education on *low-wage* workers vs. *high-wage* workers can be very different.

Practical Decision Tree



Chapter 9 Summary

Specification problems and solutions:

- **Functional form:** RESET, Davidson–MacKinnon, add quadratics/logs.
- **Omitted variables:** Proxy variables, lagged dependent variables.
- **Random slopes:** OLS works (with robust SEs) for the average effect.
- **Measurement error in y:** OLS unbiased; just larger SEs.
- **Measurement error in x (CEV):** attenuation bias toward zero.

Data problems:

- **Missing data (MCAR):** complete cases works; MIM is risky.
- **Exogenous selection:** OLS unbiased.
- **Endogenous selection:** OLS biased — needs Heckman or similar.
- **Outliers:** Detect with studentized residuals; use logs or LAD for robustness.

Looking Ahead

Limitation of Chapter 9: All solutions here use **OLS on a single cross-section**. Some problems (especially measurement error in x, endogenous selection) cannot be fully fixed this way.

Tools we will add later:

- **Time series** (Chs. 10–12): repeated observations over time.
- **Panel data** (Chs. 13–14): cross-section + time series.
- **Instrumental variables / 2SLS** (Ch. 15): direct fix for endogeneity.
- **Limited dependent variables** (Ch. 17): selection corrections, censored data.

Thank you!

Next: Chapter 10 (Time Series Basics)